



Department of Artificial Intelligence and Data Science

Course Code	AI4002
Course Title	Computer Vision
Assignment	1
Due Date	Sunday, 06 September 2026, 08:00 PM
Instructions: - Share .ipynb and .pdf file renamed as CV_A1_23F0007 - Submit task on Due Date. No Late Submission will be Tolerated.	

Question 1

Otsu's Thresholding Implementation & Analysis

I. Overview & Problem Statement

Global image thresholding assumes that an image intensity histogram is bimodal, meaning pixel values form two distinct clusters corresponding to background and foreground regions. Otsu's Method automates threshold selection by evaluating candidate split points T and selecting the threshold T^* that maximizes the between-class variance $\sigma_B^2(T)$ (or equivalently, minimizes the within-class variance $\sigma_W^2(T)$).

In this assignment, you will build an end-to-end Python pipeline that derives and applies Otsu's thresholding from first principles. You will begin by manual/programmatic tabulation on a controlled 16-level (0-15) grayscale image adapted from the lecture slides. Progress to visualization and custom binarization, and validate your custom implementation against standard library routines.

II. Input Dataset & Preprocessing

- Source Image: Use the attached image.
- Color Space: Convert the image to a single-channel grayscale representation if required.
- Quantization: Quantize the grayscale intensity values to exactly 16 intensity levels (0–15).
- Candidate Threshold Evaluation: Don't Iterate through $T \in \{0,1,2,\dots,15\}$, your evaluation must cover all 16 candidate levels.

III. Task Specifications

- Load and display the original image alongside its 16-level quantized version.
- Construct a full candidate threshold evaluation table for $T = 0,1,\dots,15$.
- Tabulate the following metrics per row:

- Threshold T
- Background weight w_1
- Foreground weight w_2
- Background mean μ_1
- Foreground mean μ_2
- Background variance σ_1^2
- Foreground variance σ_2^2
- Within-class variance σ_W^2
- Between-class variance σ_B^2

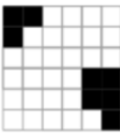
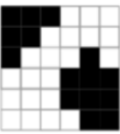
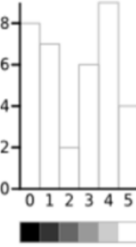
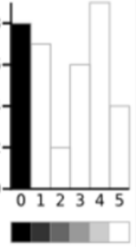
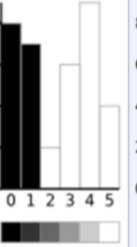
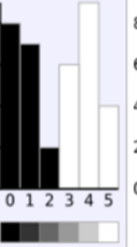
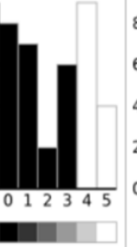
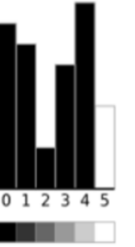
- Formally state the optimal threshold T^* and confirm whether $\max(\sigma_B^2)$ aligns with $\min(\sigma_W^2)$.
- Segment the 16-level quantized image using the binary mapping rule:

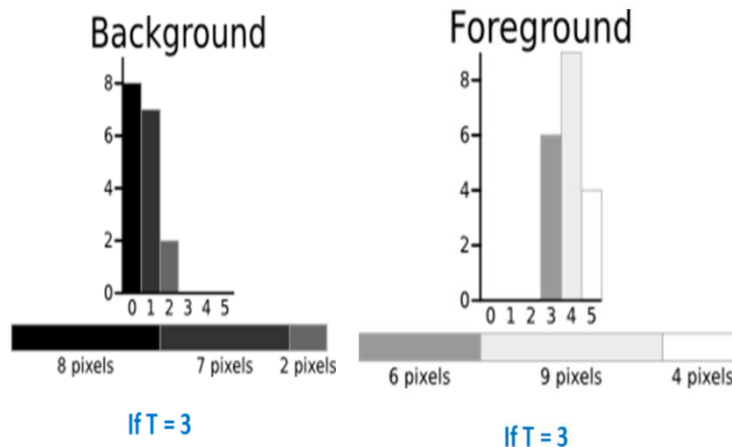
$$g(x, y) = \begin{cases} 1 & \text{if } f(x, y) > T^* \\ 0 & \text{otherwise} \end{cases}$$

- Run OpenCV (cv2.threshold with cv2.THRESH_OTSU) or Scikit-Image (skimage.filters.threshold_otsu) on the quantized image.
- Report:
 - Custom threshold vs. Library threshold.
 - Exact match confirmation (Boolean).
 - Percentage of matching pixels between custom and library binary outputs.

IV. Expected Output

Given a normalized 16-bin histogram p_i for $i \in \{0, 1, \dots, 15\}$:

Threshold	T=0	T=1	T=2	T=3	T=4	T=5
						
						
Weight, Background	$W_b = 0$	$W_b = 0.222$	$W_b = 0.4167$	$W_b = 0.4722$	$W_b = 0.6389$	$W_b = 0.8889$
Mean, Background	$M_b = 0$	$M_b = 0$	$M_b = 0.4667$	$M_b = 0.6471$	$M_b = 1.2609$	$M_b = 2.0313$
Variance, Background	$\sigma_b^2 = 0$	$\sigma_b^2 = 0$	$\sigma_b^2 = 0.2489$	$\sigma_b^2 = 0.4637$	$\sigma_b^2 = 1.4102$	$\sigma_b^2 = 2.5303$
Weight, Foreground	$W_f = 1$	$W_f = 0.7778$	$W_f = 0.5833$	$W_f = 0.5278$	$W_f = 0.3611$	$W_f = 0.1111$
Mean, Foreground	$M_f = 2.3611$	$M_f = 3.0357$	$M_f = 3.7143$	$M_f = 3.8947$	$M_f = 4.3077$	$M_f = 5.000$
Variance, Foreground	$\sigma_f^2 = 3.1196$	$\sigma_f^2 = 1.9639$	$\sigma_f^2 = 0.7755$	$\sigma_f^2 = 0.5152$	$\sigma_f^2 = 0.2130$	$\sigma_f^2 = 0$
Within Class Variance	$\sigma_W^2 = 3.1196$	$\sigma_W^2 = 1.5268$	$\sigma_W^2 = 0.5561$	$\sigma_W^2 = 0.4909$	$\sigma_W^2 = 0.9779$	$\sigma_W^2 = 2.2491$



Question 2

Intensity Transformations for Enhancement of an X-ray Image

I. Overview & Problem Statement

A medical imaging system has captured a grayscale image of an anatomical region. Due to poor acquisition conditions, the resulting image is dark and has low visual contrast. Important structures within the image are present, but their intensity values are concentrated toward the lower end of the available grayscale range, making some structures difficult to distinguish.

In medical image processing, intensity transformations can be used to modify the gray-level values of individual pixels in order to improve the visual representation of an image.

The objective is to implement these transformations manually and investigate which transformation is most appropriate for enhancing the given dark, low-contrast medical image.

II. Input Dataset & Preprocessing

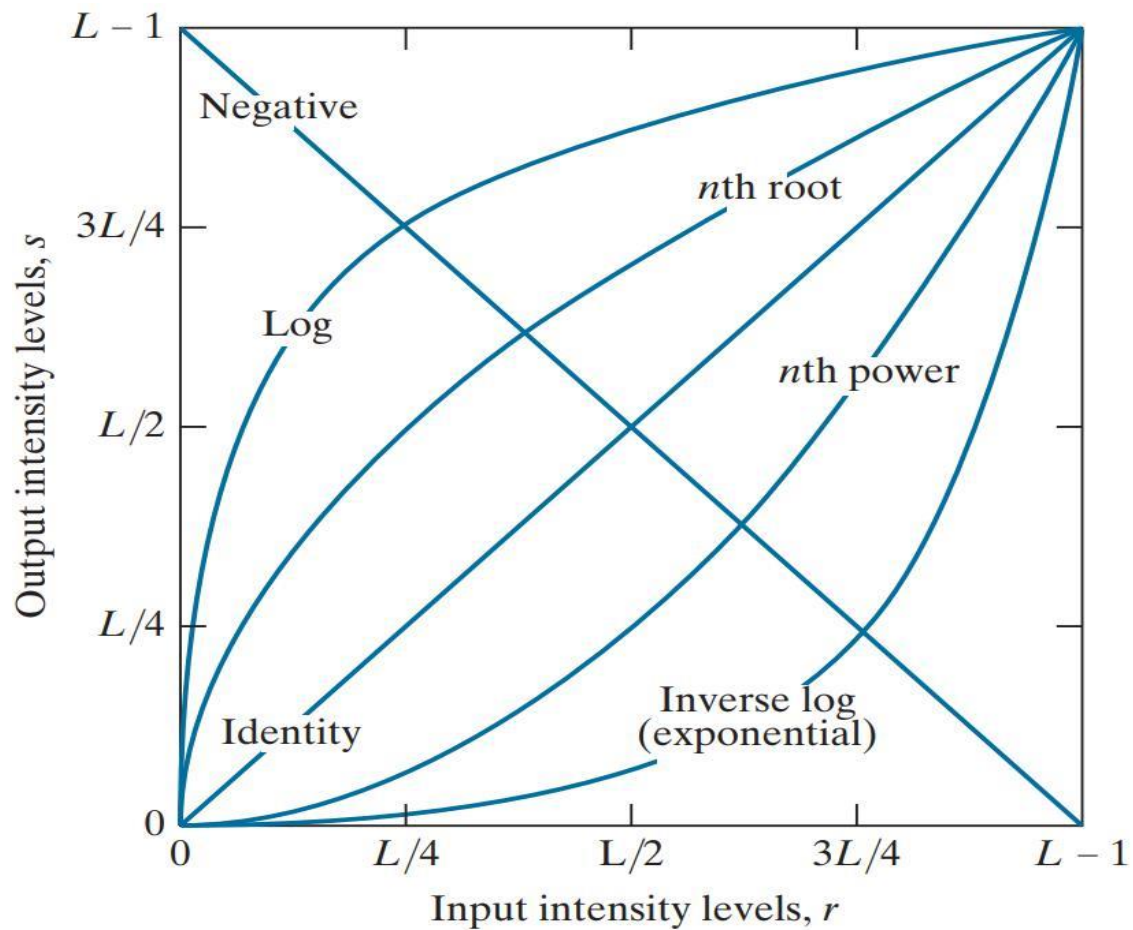
- Source Image: Use the attached image.
- Color Space: Convert the image to a single-channel grayscale representation if required.

III. Task Specifications

- Load and display the provided dark grayscale medical image and its grayscale histogram.
- Calculate and report:
 1. Minimum intensity
 2. Maximum intensity
 3. Mean intensity
 4. Standard deviation
- Implement the following intensity transformations from the graph using NumPy and display:
 1. Logarithmic transformation.
 2. Power-law transformations using multiple values of γ .
 3. Inverse-logarithmic (exponential) transformation.
- Display each transformed image along with its corresponding histogram.
- Compare the transformations based on:
 1. Brightness
 2. Contrast
 3. Intensity distribution
 4. Visibility of dark regions
- Select the single best transformation for the given dark medical image based on the visual and quantitative results.
- Justify the selected transformation by comparing its enhancement, intensity distribution, and visibility of dark regions with the other transformations.
- Implementation Constraint: All transformations must be implemented manually using NumPy. Do not use pre-built image-enhancement functions that directly perform these transformations.

IV. Expected Output

The provided graph shows several fundamental intensity transformations and their effects on input and output gray levels.



Intensity Transformation: Results

